

After we re-run the pipeline of Jenkins:

The screenshot shows the Jenkins web interface at `localhost:8080/job/my-pipeline/`. The left sidebar contains navigation links: Status, Configure, Scan Multibranch Pipeline Now, Scan Multibranch Pipeline Log, Multibranch Pipeline Events, Delete Multibranch Pipeline, Build History, Rename, Pipeline Syntax, and Credentials. The main content area displays the 'my-pipeline' job with a 'Branches (1)' section. A table lists the branches:

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☀	dev	6 min 41 sec #1	N/A	26 sec

Below the table, there is a section for 'Icon: S M L' and a 'Build Queue' section stating 'No builds in the queue.' At the bottom, the 'Build Executor Status' section shows '(0 of 2 executors busy)'.

The screenshot shows the Jenkins web interface at `localhost:8080/job/my-pipeline/job/dev/`. The left sidebar contains navigation links: Status, Changes, Build Now, View Configuration, Stages, Pipeline Syntax, and Credentials. The main content area displays the 'dev' branch with a green checkmark icon. Below the branch name, it shows the 'Full project name: my-pipeline/dev' and a 'Permalinks' section with the following links:

- [Last build \(#1\), 7 min 21 sec ago](#)
- [Last stable build \(#1\), 7 min 21 sec ago](#)
- [Last successful build \(#1\), 7 min 21 sec ago](#)
- [Last completed build \(#1\), 7 min 21 sec ago](#)

At the bottom, the 'Builds' section shows a list of builds with a search filter and a dropdown menu.

localhost:8080/job/my-pipeline/job/dev/lastCompletedBuild/

Jenkins / my-pipeline / dev / #1

Status #1 (Nov 3, 2025, 10:57:48 AM) Add description Keep this build forever

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Branch indexing

This run spent:

- 6.6 sec waiting:
- 26 sec build duration:
- 33 sec total from scheduled to completion.

git Revision: e65a96255bce7dcf692956f884deb45cef25a422
Repository: https://github.com/gaaferHajji2/j-l-jenkins-test.git

No changes.

Started 8 min 22 sec ago
Took 26 sec

localhost:8080/job/my-pipeline/job/dev/lastCompletedBuild/pipeline-overview/?selected-node=13

Jenkins / my-pipeline / dev / #1 / Stages

#1 Rerun

Started by branch indexing Started 8 min 52 sec ago Queued 6.6 sec Took 26 sec

Graph

Start Checkout SCM build test deploy End

build 0.26s Started 8m 43s ago Jenkins

building the application... 41ms

building the application...

Checkout SCM 11s

build 0.26s

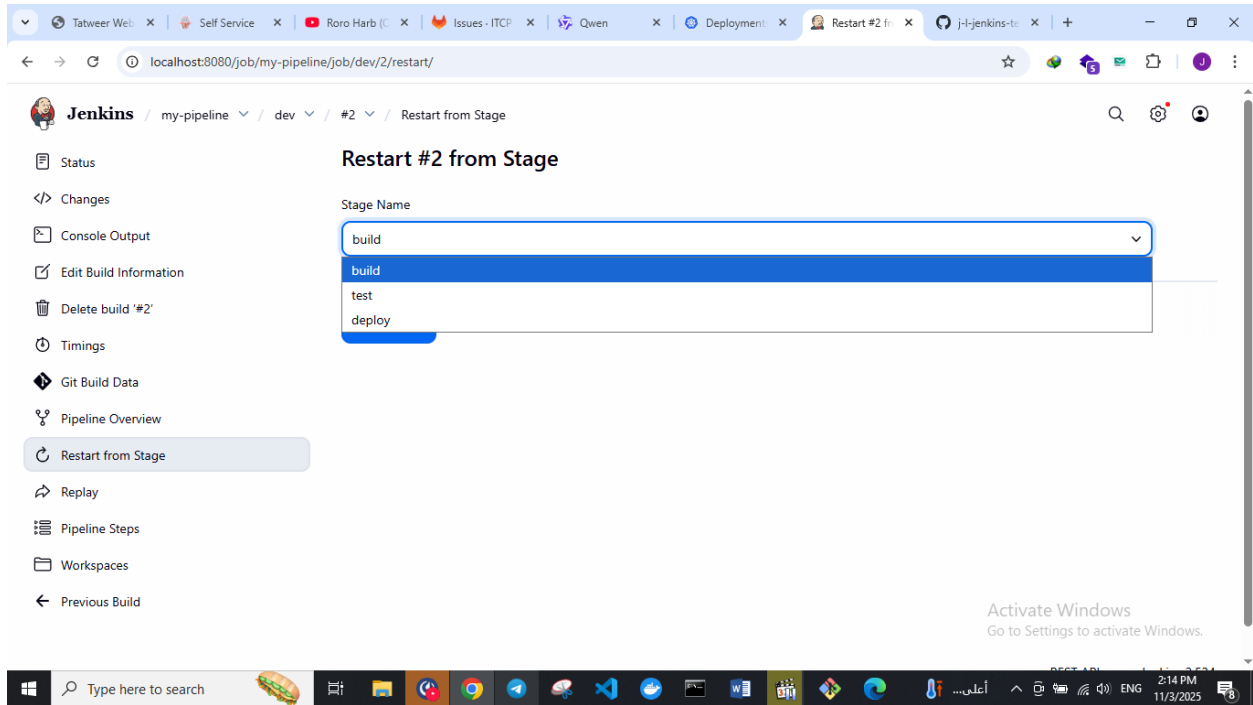
test 0.15s

deploy 0.13s

Activate Windows
Go to Settings to activate Windows.

To test changes in Jenkinsfile from Jenkins machine, we use Replay from pipeline:

The image shows two screenshots of the Jenkins web interface. The first screenshot displays build #1, which is a successful build of the 'my-pipeline' job on the 'dev' branch. The build status is 'Success' (green checkmark). The build name is '#1 (Nov 3, 2025, 10:57:48 AM)'. The build was started 13 minutes ago and took 26 seconds. The build steps include 'Branch indexing', 'This run spent' (6.6 sec waiting, 26 sec build duration, 33 sec total), 'git' (Revision: e65a96255bce7dcf692956f884deb45cef25a422, Repository: https://github.com/gaaferHajji2/j-l-jenkins-test.git, dev), and 'No changes'. The 'Replay' button is highlighted with a red arrow. The second screenshot displays build #2, which is also a successful build of the 'my-pipeline' job on the 'dev' branch. The build status is 'Success' (green checkmark). The build name is '#2 (Nov 3, 2025, 11:13:02 AM)'. The build was started 1 minute 11 seconds ago and took 9.4 seconds. The build steps include 'Started by user jenkins', 'Replayed #1 (diff)', 'This run spent' (22 ms waiting, 9.4 sec build duration, 9.4 sec total), 'git' (Revision: e65a96255bce7dcf692956f884deb45cef25a422, Repository: https://github.com/gaaferHajji2/j-l-jenkins-test.git, dev), and 'No changes'. The 'Restart from Stage' button is highlighted with a red arrow. Both screenshots show the Jenkins sidebar with options like 'Status', 'Changes', 'Console Output', 'Edit Build Information', 'Delete build', 'Timings', 'Git Build Data', 'Pipeline Overview', 'Restart from Stage', 'Replay', 'Pipeline Steps', and 'Workspaces'.



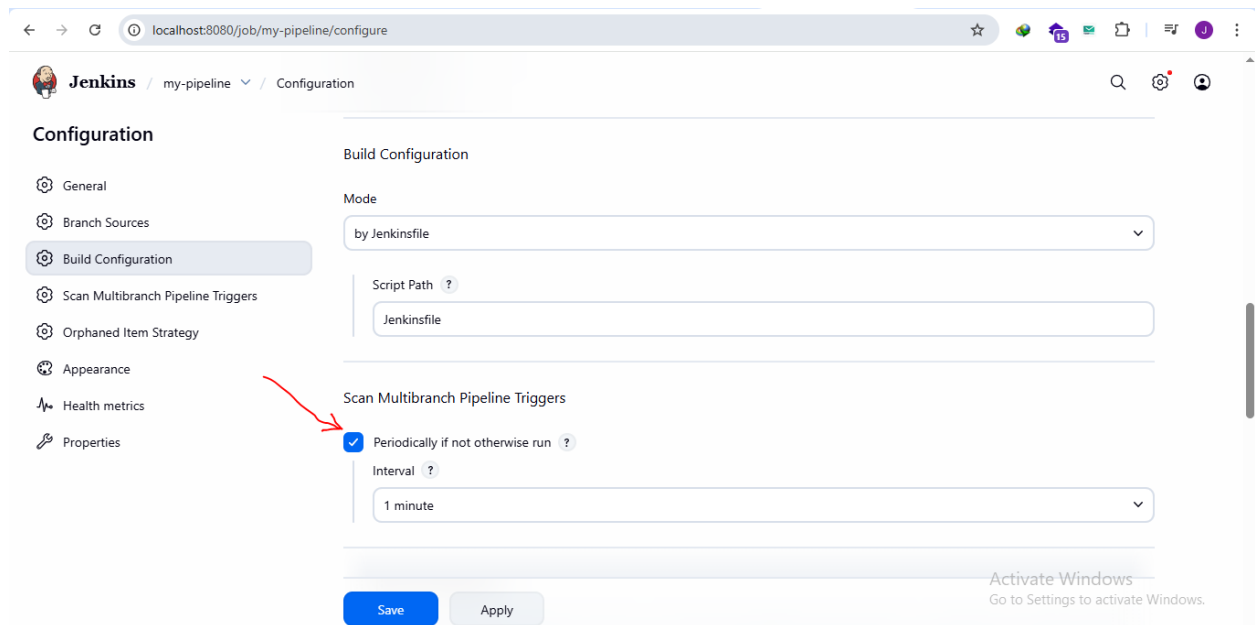
```
pipeline {  
  
  agent any  
  
  stages {  
  
    stage("build") {  
  
      steps {  
  
        echo 'building the application...'  
  
        script {  
  
          def test = 2 + 2 > 3 ? 'cool' : 'not cool'  
  
          echo test  
  
        }  
  
      }  
  
    }  
  
  }  
  
}
```

```
stage("test") {  
  
  steps {  
  
    echo 'testing the application...'  
  
  }  
  
}
```

```
stage("deploy") {  
  
  steps {  
  
    echo 'deploying the application...'  
  
  }  
  
}
```

```
}  
  
}  
  
}  
  
}
```

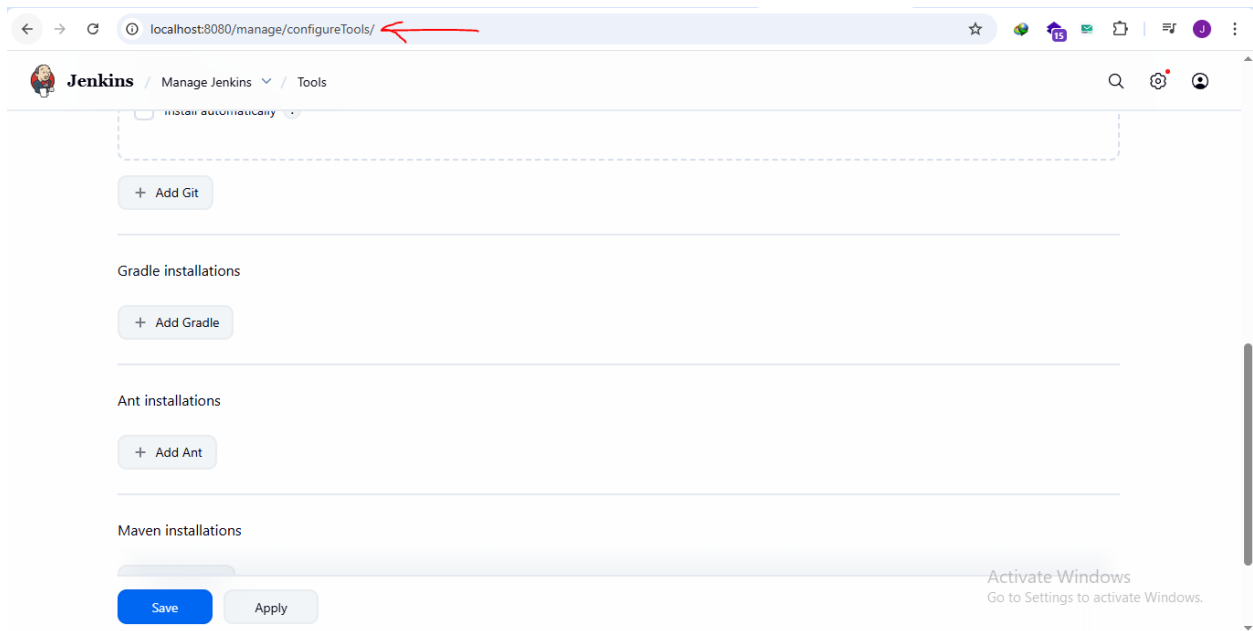
To poll the changes from Jenkins to github:



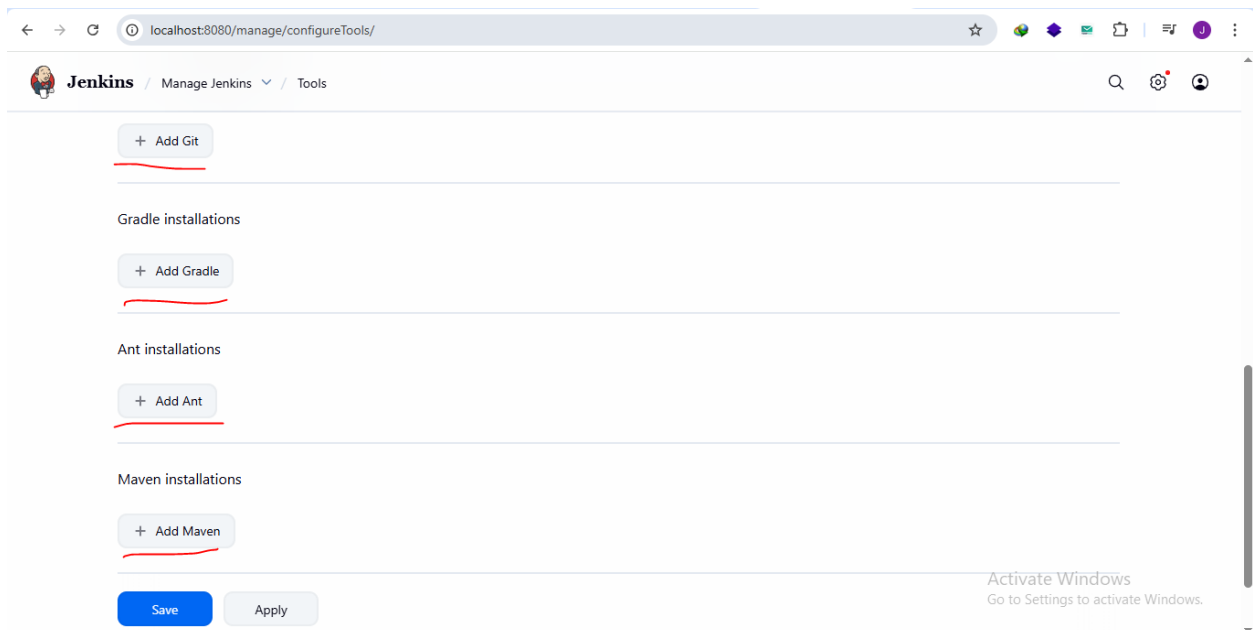
The other way to check the changes in github by using push notification, but it is not always works, especially if the Jenkins server stops

The perfect way is to use both ways (poll and push notifications)

To configure any tool in Jenkins, from manage Jenkins ➡ configure tools:



If we have btn with: Add ... ➡ that means we must configure it first



For Gradle we only need to set the name and installation version:

localhost:8080/manage/configureTools/

Jenkins / Manage Jenkins / Tools

+ Add Gradle

Gradle

name ?
Gradle

☒ Install automatically ?

Install from Gradle.org

Version
Gradle 6.2

+ Add Installer

Save Apply

Activate Windows
Go to Settings to activate Windows.

After installing node.js from plugin manager, we go to configuration to config it:

localhost:8080/manage/configureTools/

Jenkins / Manage Jenkins / Tools

☒ Install automatically ?

Install from nodejs.org

Version
NodeJS 25.1.0

For the underlying architecture, if available, force the installation of the 32bit package. Otherwise the build will fail

☐ Force 32bit architecture

Global npm packages to install
Specify list of packages to install globally -- see npm install -g. Note that you can fix the packages version by using the syntax `packageName@version`

Global npm packages refresh hours
Duration, in hours, before 2 npm cache update. Note that 0 will always update npm cache

72

Save Apply

Activate Windows
Go to Settings to activate Windows.

For 1 & 2 ➔ They are called wrappers.

For node.js build tools that we install it from plugins; it's called: nodejs

For gradle build tools that we install it from plugins; it's called: withGradle

For 3 it is the name of installation from the configuration:

```
pipeline {
  agent any
  stages {
    1 stage("run frontend") {
      steps {
        echo 'executing yarn...'
        nodejs('Node-10.17') {
          sh 'yarn install'
        }
      }
    }
    2 stage("run backend") {
      steps {
        echo 'executing gradle...'
        withGradle() {
          sh './gradlew -v'
        }
      }
    }
  }
}
```

NodeJS installations ^ Edited

+ Add NodeJS

NodeJS

Name

Node-10.17

☒ Install automatically ?

Install from nodejs.org

Version

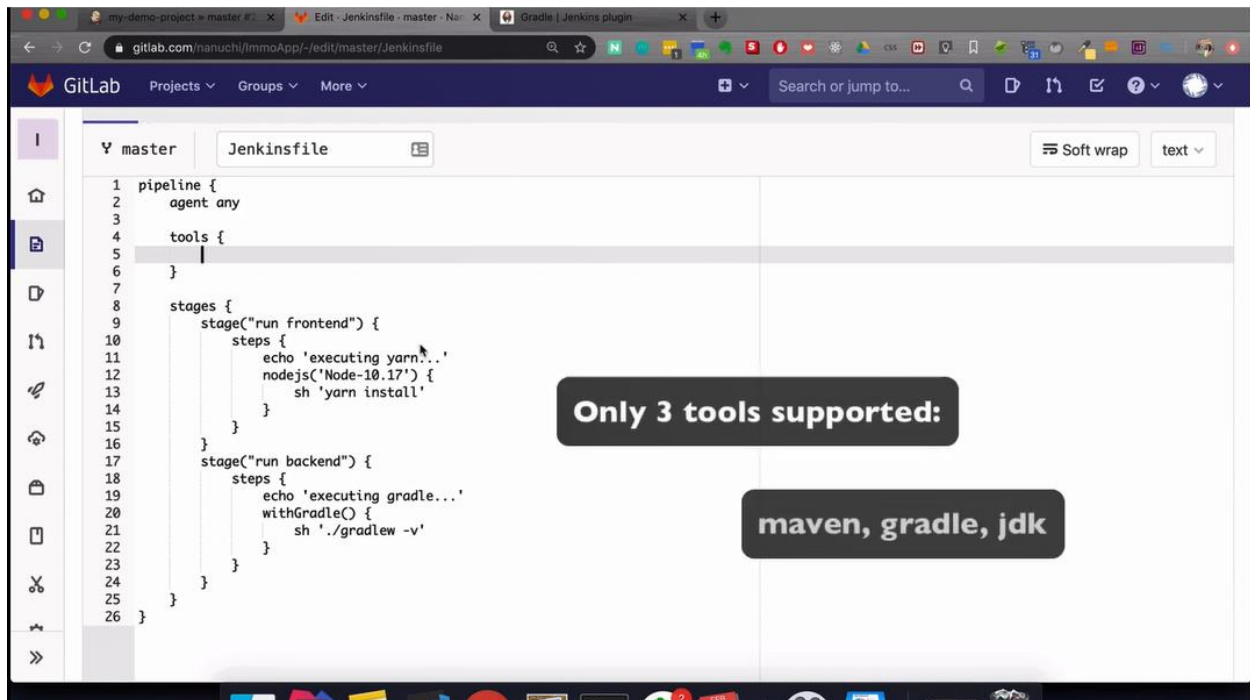
NodeJS 10.17.0

For the underlying architecture, if available, force the installation of the 32bit package. Otherwise the build will fail

Save Apply

Activate Windows
Go to Settings to activate Windows.

To build without wrappers, we use tools:



```
pipeline {
  agent any

  tools {
    gradle 'Gradle-6.2'
  }

  stages {
    stage("run frontend") {
      steps {
        echo 'executing yarn...'
        nodejs('Node-10.17') {
          sh 'yarn install'
        }
      }
    }
    stage("run backend") {
      steps {
        echo 'executing gradle...'
        sh './gradlew -v'
      }
    }
  }
}
```
